

Artificial Intelligence and Quantum Computing- A Symbiotic Loop?

What is an AI – Led Economy?

The way I think about an AI – Led Economy, is a world where the primary human contribution gradually shifts from manual brut physical effort and step-by-step execution towards human intent, judgement, creativity and direction.

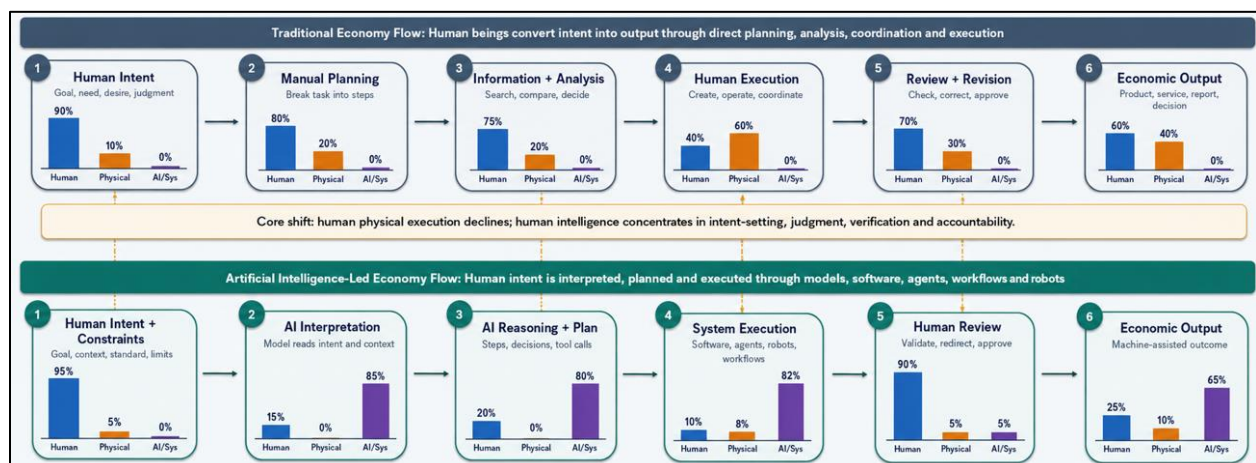
In a traditional economy, economic output often depends on how much human effort is applied directly to a task. A person must plan the work, gather information, analyze it, make decisions, create the output, coordinate execution, and revise the result.

In an Artificial Intelligence-Led Economy, the starting point increasingly becomes human intent.

The primary change is that humans move higher in the value chain. Instead of spending most of their time performing repetitive execution, they spend more time defining the right problem, asking the right question, setting the right constraints, judging the quality of the output, and deciding what should be done next.

Thus, the quality of the output follows the quality of the human intent. I am considering the underlying questions, right constraints and judgement as part of the human intent.

Below is a diagram which illustrates the AI-Led economy which I imagine and how its operating model differs from a traditional economy.:



Artificial Intelligence-Led Economy Becomes a Physical Infrastructure Problem and Hence the famous Capex Cycle

Underneath the beautiful AI Led economy sits a monstrous physical industrial system. When human intent is interpreted by artificial intelligence models and converted into execution, the work still must be performed somewhere. That “somewhere” is not abstract. It is a chain of chips, high-bandwidth memory, servers, storage, optical networking, data centers, power equipment, cooling systems, land, water, and electricity.

In the case of traditional economy, the primary bottlenecks were – Human bandwidths: time, skill, coordination, judgement and physical effort. In an AI-Led economy, significant part of these bottlenecks shifts towards physical infrastructure.

Human intent becomes machine reasoning → machine reasoning becomes compute demand → compute demand becomes electricity consumption → electricity consumption becomes heat → heat requires cooling, power distribution, and physical facilities.

Thus, artificial intelligence does not eliminate the need for physical input. It changes where the physical burden sits.

The International Energy Agency estimates that global data centers consumed about 415 terawatt-hours of electricity in 2024, around 1.5% of global electricity use. By 2030, it expects data-center electricity consumption to more than double to about 945 terawatt-hours, or just under 3% of global electricity demand. It also estimates that from 2024 to 2030, data-center electricity consumption will grow by around 15% per year, much faster than electricity demand growth in most other sectors. (source: <https://www.iea.org>)

The United States Department of Energy summarized the Lawrence Berkeley National Laboratory’s 2024 report by saying that United States data centers consumed about 4.4% of total United States electricity in 2023 and could consume roughly 6.7% to 12% by 2028. The same report estimated that United States data-center electricity use rose from 58 terawatt-hours in 2014 to 176 terawatt-hours in 2023 and could reach 325 to 580 terawatt-hours by 2028. (source: <https://www.energy.gov/articles/doe-releases-new-report-evaluating-increase-electricity-demand-data-centers?>)

Biggest infrastructure bottlenecks in the Artificial Intelligence-Led Economy

We can rank the bottlenecks for AI-Led Economy as below:

Rank	Bottleneck	Estimated weight
1	Power availability and grid interconnection	18%
2	Graphics processing unit / accelerator supply	16%

Rank	Bottleneck	Estimated weight
3	Memory bandwidth and high-bandwidth memory	13%
4	Heat removal and liquid cooling	11%
5	Networking bandwidth and latency	10%
6	Data-center site, land, permits, and construction	9%
7	Electrical equipment: transformers, switchgear, uninterruptible power supply	8%
8	Advanced packaging and chip yield	6%
9	Software efficiency and model optimization	5%
10	Water, sustainability, and local community constraints	4%

Where Artificial Intelligence capital expenditure is flowing

A useful normalized view of a 100-unit Artificial Intelligence infrastructure capital expenditure budget would look like this:

Rank	Capital expenditure area	Estimated share
1	Accelerators, servers, and compute racks	32%
2	Data-center construction and fit-out	14%
3	Power generation, procurement, and grid interconnection	12%
4	Electrical equipment and power distribution	10%
5	High-bandwidth memory and storage systems	9%
6	Networking switches, optics, and interconnects	8%
7	Cooling and thermal management	6%
8	Advanced packaging, testing, and semiconductor equipment	4%
9	Security, orchestration, monitoring, and data-center software	3%
10	Redundancy, backup systems, and resilience	2%

How AI capex complements Quantum computing infrastructure:

For a normalized 100-Unit Artificial Intelligence infrastructure capital expenditure budget, the percentage distribution of capex and its adjacency with Quantum computing infrastructure requirement, can be summarized as below:

AI capex area	AI capex share	Adjacency to quantum capex	Rationale
Accelerators, servers, compute racks	32%	70%	Quantum will need classical high-performance computing for simulation, control, and error correction. NVIDIA says useful quantum processors must work with artificial intelligence supercomputers. (NVIDIA)
Data-center construction and fit-out	14%	55%	Quantum systems also need controlled facilities, uptime, security, and integration into cloud or high-performance computing environments.
Power procurement and grid interconnection	12%	45%	Quantum systems are not as power-hungry as artificial intelligence data centers today, but cryogenics, control electronics, and classical servers need highly reliable power.
Electrical equipment and power distribution	10%	65%	Both artificial intelligence and quantum systems need stable power delivery, redundancy, and specialized electrical design.
High-bandwidth memory and storage	9%	35%	Quantum processors do not use memory like graphics processing units, but quantum simulation, compilation, and error-correction workloads still need classical data systems.
Networking, switches, optics, interconnects	8%	75%	Quantum-classical workflows require low-latency links. NVIDIA's NVQLink is designed to connect quantum hardware with accelerated computing for calibration and quantum error correction. (NVIDIA)
Cooling and thermal management	6%	85%	Artificial intelligence uses liquid cooling; quantum systems require cryogenic cooling. The methods differ, but both depend on advanced thermal engineering. IBM says its quantum systems require ultra-cold cryogenic infrastructure. (ibm.com)

AI capex area	AI capex share	Adjacency to quantum capex	Rationale
Advanced packaging, fabrication, testing	4%	80%	IBM says its qubits use state-of-the-art 300-millimeter semiconductor chip fabrication and advanced packaging methods. (ibm.com)
Software orchestration and monitoring	3%	90%	Quantum will need hybrid workflows, middleware, control software, and cloud orchestration. IBM describes quantum-centric supercomputing as requiring software and middleware for hybrid quantum-classical workflows. (IBM Research)
Resilience, backup, security	2%	60%	Both infrastructures need cybersecurity, uptime, identity, governance, and operational resilience.

Now, if we apply **weighted adjacency formula** as = (AI capex share in each area) × (Quantum adjacency score of that area), and add all the results, we will find roughly **63% of Artificial Intelligence infrastructure capital expenditure has medium-to-high adjacency with future quantum computing capital expenditure**.

This does not mean quantum will reuse artificial intelligence infrastructure one-for-one. Quantum needs specialized cryogenics, qubit control, lasers or microwave systems, error-correction hardware, and quantum processors. But the artificial intelligence buildout creates adjacent rails: cloud data centers, power systems, advanced networking, semiconductor fabrication, control software, cooling know-how, and high-performance computing.

That is why Artificial Intelligence capital expenditure may become complementary to quantum capital expenditure: artificial intelligence builds the industrial base, while quantum may later attach to that base as a specialized accelerator for simulation, optimization, sampling, materials discovery, and energy-system design.

Increasing need for intelligent Power Optimization

As the old digital economy, which broadly consisted of **search, retrieval, display, storage, transmission and routing of information**, changes into an Artificial Intelligence-led economy, the operating model also changes. At a high level, the new flow becomes **Intent → Model Inference → Generate / Summarize / Recommend / Decide → Orchestrate → Output or Action**.

In such an economy, millions of human-generated and agent-generated questions, prompts and task requests convert into recurring inference demand. This inference demand then becomes computed demand. Thus, intelligence-based economic activity in an Artificial Intelligence-led economy needs the

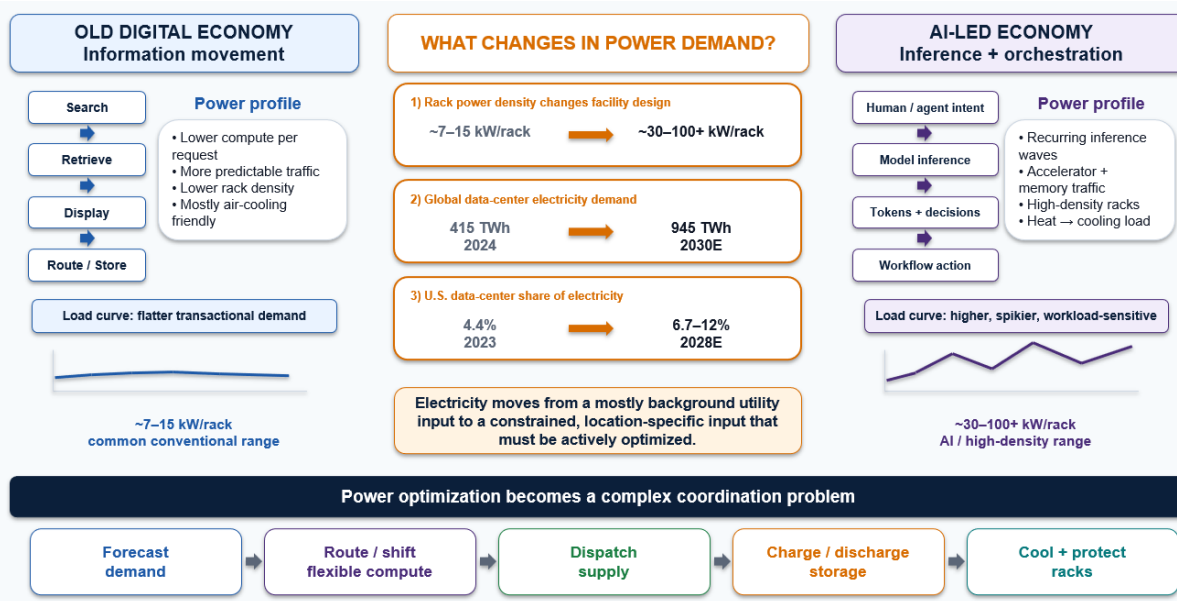
support of more models, more model calls, more tokens, more memory movement, and more concurrent inference sessions.

Increased inference computes raise the need for higher data movement inside and across data-center clusters: accelerator-to-accelerator traffic, rack-to-rack traffic, storage-to-compute traffic, data-center-to-data-center routing, and in some cases edge inference on laptops, mobile devices, vehicles and Internet of Things devices. At the same time, more compute activity increases electricity demand inside the data-center rack, from power supply units to busbars, voltage-regulator modules, processors, memory systems, networking equipment and cooling systems.

Thus, as the Artificial Intelligence-led economy expands and diffuses across daily economic activity, increased inference load translates into increased demand for deliverable power.

However, the problem is not merely that electricity demand increases. The deeper problem is that deliverable power is finite at a specific location and at a time. Power must now be coordinated more intelligently among residential homes, public infrastructure, commercial activity, industrial users and Artificial Intelligence data centers. This creates the need for intelligent power optimization across generation, transmission, substations, storage, data-center power infrastructure, cooling systems and flexible compute workloads.

Thus, eventually in an AI-Led economy the need to efficiently and intelligently decide where power should flow, when flexible workloads should move or pause, when batteries should discharge, when backup systems should remain on standby, and how critical human and economic activity should be balanced becomes a more complex and important problem to solve.



Sources/assumptions: IEA Energy and AI 2025; Lawrence Berkeley National Laboratory / U.S. Department of Energy 2024 data-center report; rack-density ranges are directional industry estimates. Numbers describe data-center electricity demand, not total economy-wide electricity demand.

Why is the Hardest Edge Intelligent Power Optimization problem in an AI-Led Economy a Quantum Problem?

At the hardest edge, this becomes a high-dimensional decision problem.

Quantum computing becomes relevant when four conditions appear together. **One condition alone is not enough.**

1. Massive combinatorial complexity

In Quantum language - combinatorial optimization, Quadratic Unconstrained Binary Optimization, Ising formulation, search-space explosion and candidate-state sampling.

What it basically means is many binary and discrete decisions must be evaluated together - which generators commit, which storage assets charge or discharge, which transmission paths stay open or constrained, which Artificial Intelligence workloads run now, pause or move to another site, and which backup units remain reserved.

As assets increase, the number of combinations grows non-linearly. This is the part that can be mapped into Quadratic Unconstrained Binary Optimization or Ising-type structures when variables are simplified into binary decisions and penalties.

2. Strong interdependence across assets

In Quantum language - coupling terms, interaction graph and cost Hamiltonian.

It simply means each decision changes the value and feasibility of other decisions. Starting a gas turbine changes available reserves. Moving Artificial Intelligence workload from Site A to Site B changes network load, cooling load and local power draw. Discharging a battery changes near-term reliability and future optionality. Cooling decisions affect thermal headroom and power usage. These are not independent switches. They are coupled variables in a networked objective function.

3. Uncertainty and fast-changing conditions

In Quantum language - scenario sampling, probabilistic state distribution and hybrid quantum-classical re-optimization.

Which means the system is not static. Renewable generation changes, grid congestion changes, inference demand changes, weather changes, equipment availability changes and market prices change. The optimization engine must repeatedly evaluate possible futures based on current scenarios, not only fixed forecasts using past data.

4. Tight pressure and low tolerance for faulty decisions

In Quantum language - variational optimization, Quantum Approximate Optimization Algorithm, quantum annealing and approximate feasible solutions.

The operator does not need a mathematically perfect answer after the crisis has passed. The operator needs a high-quality feasible decision fast enough to matter.

This is why the hardest-edge intelligent power-optimization problem becomes quantum-relevant. It asks the system to choose among many discrete actions, respect many physical and economic constraints, and continuously update the answer under uncertainty and time pressure in real-time.

In real life, the decision space includes:

- Which generators should run gas, nuclear, fossil fuel, hydro, wind, solar or distributed backup.
- How power should be routed through transmission lines, substations and transformers.
- When batteries should charge, discharge or remain idle.
- How cooling systems should respond without damaging data-center uptime.
- Whether Artificial Intelligence workloads should run now, defer, throttle or shift from Site A to Site B.
- Whether backup power should be committed or preserved as reserve.

The constraints are also simultaneous:

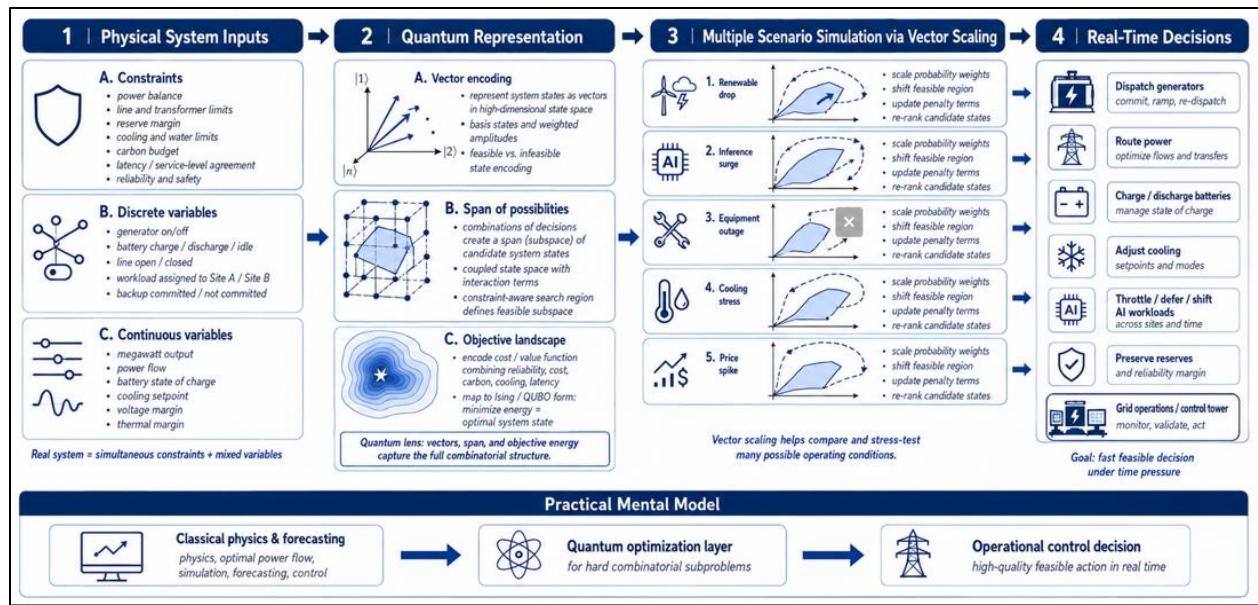
- power balance
- transmission-line capacity
- transformer and thermal limits
- reserve margin
- cooling and water limits
- carbon-Emissions budget
- service-level agreement latency
- reliability and safety limits

The variables are mixed.

- ⇒ Some are discrete: generator on/off, line open/closed, battery charge/discharge/idle, workload assigned to Site A or Site B, and backup unit committed or not.
- ⇒ Others are continuous: generator output in megawatts, power flow in megawatts, battery state of charge, cooling setpoint, voltage margin and thermal margin.

Classical optimization already solves many versions of these problems today using mixed-integer programming, optimal power flow, heuristics, simulation and real-time control. So, the mental model is definitely not that quantum computers will replace classical computers tomorrow.

The more precise mental model is this: as the Artificial Intelligence-led economy adds flexible compute load, distributed energy resources, storage, cooling constraints, carbon targets and real-time inference demand, the optimization problem becomes more combinatorial, more coupled, more probabilistic and more time-sensitive. That is the zone where quantum optimization methods are being conceptualized as an augmentation factor for classical compute solutions, which would still handle 80%-90% of the scenarios.



IonQ and Oak Ridge National Laboratory partnership to solve Unit Commitment Problem

IonQ is a publicly listed company which develops trapped ion quantum computing systems. It provides access to its system over cloud (Azure, Google Cloud or AWS), and helps enterprises to research quantum solutions for real world computational problems.

Oak Ridge National Laboratory is a United States Department of Energy lab focused on science, energy, national security, supercomputing, and advanced research.

IonQ and Oak Ridge National Laboratory have recently announced partnership to attack the **Unit Commitment** problem in power-grid operations

What is a **Unit Commitment** problem in power-grid operations => Deciding which generators should be switched on or off across time, and how much power each committed generator should produce, while satisfying hourly demand and minimizing cost. Technically, this is a **mixed-integer nonlinear optimization** problem: generator commitment is binary/discrete, while generator output is continuous (<https://arxiv.org/abs/2505.00145>)

The scale problem is mathematical. If there are N generators and M time periods, the possible commitment schedules grow as $(2^N - 1)^M$. The IonQ–Oak Ridge paper gives an example of 15 generators over 24 hours, producing roughly 2.3×10^{108} possible schedules. This directly matches the earlier section's quantum lens: vectors, span, scenario explosion, discrete variables, continuous variables, and constrained search - searching for the best decision inside a huge universe of possible choices, but only among choices that obey the physical, economic, and reliability rules of the power system.

Their hybrid method uses a Variational Quantum Algorithm to generate candidate Unit Commitment vectors, and a classical optimizer to calculate the corresponding generator output levels.

Their method works like a two-step decision process:

First, the quantum computer helps explore possible on/off schedules for power generators. For example, if there are 26 generators, each generator can be either:

1 = ON

0 = OFF

So one possible schedule for a single time period may look like this:

[1, 0, 1, 1, 0, 0, 1, ...]

This is called a Unit Commitment vector. It is a compact mathematical way of saying: “These generators are running, and these generators are not.”

The hard part is that the number of possible schedules grows extremely fast. If there are **N generators** and **M time periods**, the possible schedules grow approximately as:

$$(2^N - 1)^M$$

So, with **26 generators** and **24 time periods**, the system is searching through a massive universe of possible generator schedules.

IonQ’s quantum system is used to help generate promising candidate schedules from this huge space. Then a classical computer takes those candidate schedules and calculates the practical details: how much electricity each selected generator should actually produce, while respecting demand, cost, ramping limits, and reliability rules.

Thus IonQ’s system answers the question - “**Which generators should be considered for running?**”

The classical computer then answers - “**Given that selection, exactly how much power should each generator produce?**”

In the IonQ–Oak Ridge work, this hybrid quantum-classical method was tested on systems with 3, 10, and 26 generating units, across 24 time periods. IonQ later stated that the work used its 36-qubit Forte Enterprise quantum computer.

Therefore, the IonQ–Oak Ridge case study is not an isolated quantum experiment. It is an early signal of where the grid problem is moving from static planning towards dynamic, high-dimensional, hybrid computation.

The grid is becoming more digital, more renewable, more distributed, more constrained, and more exposed to large flexible loads such as artificial intelligence data centers. That means the hardest problem is no longer only “how much power do we need?” The harder problem becomes “which combination of assets, loads, storage, cooling systems, and flexible workloads should be activated, deferred, routed, throttled, or reserved under current and possible future conditions?”

Conclusion

The relationship between Artificial Intelligence and Quantum Computing is best understood as a reinforcing capital and capability loop.

Artificial Intelligence is already forcing the world to build a much larger physical infrastructure base: chips, high-bandwidth memory, data centers, optical networks, power systems, cooling capacity, semiconductor manufacturing, and advanced control software. A meaningful portion of this Artificial Intelligence capital expenditure has adjacency with the future Quantum Computing buildout, because Quantum Computing will also depend on many of the same industrial rails, even though it will require specialized quantum processors, cryogenics, qubit-control systems, lasers or microwave systems, and error-correction hardware.

This makes Artificial Intelligence a practical indirect beneficiary to Quantum Computing infrastructure. Artificial Intelligence creates economic pull, expands the high-performance computing ecosystem, increases demand for advanced semiconductors and data-center-grade power systems, and exposes harder real-world problems that classical compute may struggle to solve efficiently at the edge of scale.

Quantum Computing, in turn, becomes relevant where the problem structure moves beyond ordinary computation into massive combinatorial complexity, coupled variables, probabilistic scenarios, constrained search, simulation, and time-sensitive optimization. The intelligent power-optimization problem in an Artificial Intelligence-led economy is a strong example: the system must decide how generators, batteries, transmission lines, cooling systems, and flexible Artificial Intelligence workloads should operate together under physical, economic, and reliability constraints.

The likely future is hybrid. Classical computers will remain the dominant workhorses for most tasks. Quantum systems may emerge as specialized accelerators for the hardest optimization, simulation, and search problems. In that sense, Artificial Intelligence builds the infrastructure and demand base; Quantum Computing adds a new computational layer for problems where scale, uncertainty, and constraint density make better search and simulation economically valuable.